

KHI Proprietary LH2 Solution vs. Public Literature — Comparison

Cross-reference of Kawasaki Heavy Industries' (KHI) proprietary LH2 offer (as disclosed to Gentari) against two peer-reviewed public papers already ingested by the research sub-agent ("Zane"). Built as a **repo-internal deliverable** in docs/comparison/ — separate from Zane's isolated staging area (research/sources/staging.csv); no staging/promotion status is changed by this file, per research/AGENT.md 's isolation policy.

Purpose. Support KR1.3 (LH2-vs-alternative-carrier comparison) by positioning KHI's disclosed engineering/commercial figures against the wider academic literature on LH2 maritime transport — where they corroborate, where they diverge, and where KHI simply has not disclosed a figure the literature does address (and vice versa).

Scope / system boundary. Full LH2 chain as far as each source discloses: production is out of scope for all three sources; liquefaction → export terminal → shipping (laden voyage) → import terminal → regasification is covered to varying depth. Where a source's boundary differs (e.g. "voyage" vs "terminal at rest", or "transportation cost" vs "full chain LCOH"), this is called out explicitly rather than merged.

Units & cost-year. SI units (kg, %/day, K, bar, m³, MJ/kg, kWh/kg) per docs/conventions.md . Costs are shown in their **native, disclosed unit** (the IAE stack is unlabelled-axis JPY/Nm³-equivalent, ~2019 vintage; the RSER figure is USD/GJ, cost-year not stated in the paper) — units are **not** forced onto a common basis here because the underlying cost bases differ (see §5).

Provenance-separation rule. Every cell is tagged with its source provenance: - **[KHI]** — Kawasaki Heavy Industries, proprietary, Gentari-confidential (do not redistribute outside Gentari/KHI engagement). - **[RSER]** — Towhid & Hossain, *Renewable & Sustainable Energy Reviews* 233 (2026) 116850 (public, open literature). - **[JMSE]** — Passalacqua & Traverso, *J. Mar. Sci. Eng.* 2025, 13(9), 1748 (public, open-access literature).

No cell blends KHI and paper figures into one unattributed number. Where KHI discloses no figure, the cell says so explicitly rather than being left blank or inferred from the papers.

Sources used (citation ids): - kawasaki-2026-comparison , kawasaki-2026-supplemental , kawasaki-2026-questionnaire , kawasaki-hydrogen-activities , iae-2019-gigaton — all in data/references.csv (proprietary tier). - rser-2026-lh2-maritime-review — in research/sources/staging.csv (journal tier); full text verified at research/literature/rser-2026-lh2-maritime-transportation.fulltext.md . - jmse-2025-lng-to-lh2-review — in research/sources/staging.csv (journal tier); full text verified at research/literature/jmse-2025-lng-to-lh2-maritime-review.fulltext.md .

Every paper figure below was re-verified against the actual .fulltext.md (not just the staging.csv summary/abstract) before being cited; page/line context is noted in the notes column where it materially affects interpretation (e.g. whether a figure is the paper's own result or a secondary source the paper is reviewing).

1. Boil-off rate (BOR) — at-rest vs voyage

Metric	[KHI]	[RSER]	[JMSE]
At-rest terminal BOR	0.1 %/day , both loading and receiving terminal tanks (KHI, 2026, kawasaki-2026-supplemental)	Not the paper's focus (paper is voyage-centric); no at-rest terminal figure independently modelled	Not quantified for a specific terminal; discusses BOR sensitivity qualitatively (see below)
Voyage/laden BOR (commercial-scale ship)	Not disclosed — KHI. KHI states only that "the consumption rate of fuel gas (H ₂) is equal to BOR" and gives no standalone %/day figure for the 160,000 m ³ commercial vessel (KHI, 2026, kawasaki-2026-questionnaire , Q23)	~3.44 %/day , for a 160,000 m ³ LH2 carrier, 4 spherical tanks (radius 21.23 m), 30-day baseline voyage, passive insulation ~0.1 W/m ² K (Towhid & Hossain, 2026, fulltext lines 638–668, 1370–1375). Note: this is RSER's synthesis of a cited technical-analysis study within the review (not the RSER authors' own new experiment) — RSER attributes the ~3.438%/day figure to that underlying study's sensitivity analysis across carriers	Not modelled as a standalone %/day figure for a commercial-scale ship; but reports an observed real-world data point : on the Suiso Frontier's demonstration voyage (Australia → Japan, 1,250 m ³ pilot vessel), "roughly 10% of the loaded LH2 is dispersed through the vent mast" over the trip (Passalacqua & Traverso, 2025, fulltext lines 405–409)
Tank-diameter scaling of BOR	Not addressed — KHI's disclosed figures are point values for its own 64,000–65,000 m ³ /160,000 m ³ scale, no scaling curve given	Larger tank diameter reduces relative BOG losses from ~1.8 %/day (small, ~2 m³) to ~0.2 %/day for tanks up to ~1,200 m ³ , "due to thermal inertia," validated against NASA's MHTB experimental tank data (Wang et al. 2024, as reviewed in Towhid & Hossain 2026, fulltext lines 1082–1083, 1370–1372). Note: this scaling study covers 2–1,200 m³ tanks — two to three orders of magnitude smaller than KHI's 64,000–160,000 m³ tanks; extrapolating the curve to KHI's scale is not something either source supports	Not quantified as a diameter-scaling curve; states qualitatively that LH2 BOR (unlike LNG's) is "strongly affected" by ambient temperature and storage size (fulltext lines 359–362)
Single-node vs multi-zone modelling error	Not applicable — KHI does not publish a thermodynamic model	Single-node equilibrium models can under-predict internal tank-wall energy by up to 60% at low fill levels (≤5%), when max vapour temperature exceeds ~100 K (Wang et al. 2024, as reviewed, fulltext lines 1091–1100, 1108)	Not addressed at this level of modelling detail
BOG energy-recovery potential (bulk carrier)	Not addressed	Not this specific figure	BOG treatment/utilization can recover ~4–6% of the energy otherwise lost through spontaneous evaporation, for large-volume LH2 bulk carriers (as opposed to fuel-tank scale) (fulltext lines 363–367)

Read: KHI's only hard, disclosed BOR number is the 0.1%/day at-rest terminal figure; it has never given a standalone voyage BOR for its commercial 160,000 m³ ship concept. The literature's ~3.44%/day figure is at a comparable ship scale (160,000 m³) but is itself a secondary citation within RSER, not a first-principles KHI-validated number — it should be treated as an independent modelling estimate, not a benchmark KHI has endorsed. The one real *operational* data point available anywhere in these three sources is JMSE's ~10% total voyage loss on the 1,250 m³ Suiso Frontier demonstration run — but that is a small pilot vessel, not representative of the 160,000 m³ target scale.

2. Tank design / insulation

Metric	[KHI]	[RSER]	[JMSE]
Tank architecture (export/loading)	64,000 m³/tank , 3 tanks (Base, 2.5 Bn Nm³/y) or 4 tanks (Large, 10 Bn Nm³/y) (KHI, 2026, kawasaki-2026-supplemental)	N/A (not a design-taxonomy paper for terminal tanks)	N/A
Tank architecture (receiving)	65,000 m³/tank , 3 tanks (Base) or 11 tanks (Large) (KHI, 2026, kawasaki-2026-supplemental)	N/A	N/A
Onboard/vessel tank type	Commercial target vessel 160,000 m³ , 2 ships (Base)/8 ships (Large); construction underway for a 40,000 m³ vessel using "the same technology of cargo tanks proven by Suiso Frontier" (KHI, 2026, kawasaki-2026-supplemental , -questionnaire Q29). Design description: " double-shelled/vacuum insulated tank ", "high-performance insulation technology" (KHI, kawasaki-hydrogen-activities , company-level, 2024, not project-specific). One option raised for ground tanks — but not confirmed as the chosen design — is a spherical tank with pressurized send-out to minimize dead volume (KHI, 2026, kawasaki-2026-questionnaire Q4)	Reviews an IMO Type-taxonomy framework generically (Type A prismatic, Type B spherical/Moss or prismatic, Type C cylindrical pressurized) developed for LNG , and notes spherical tanks (Type B, Moss-type) "showed the lowest BOG rate" in a cited thermodynamic sensitivity study because of minimum surface-area-to-volume ratio (fulltext lines 638–650); does not assign an IMO type specifically to LH2 carriers	Same IMO Type A/B/C taxonomy (fulltext lines 171–182), explicitly for LNG ; separately confirms the Suiso Frontier LH2 tanker "both the inner vessel and the outer containment are realized utilizing austenitic stainless steel" (fulltext lines 461–464) with glass-fibre-reinforced plastic (GFRP) used for the inner-vessel saddles (fulltext lines 473–475). LH2 tank & insulation TRL: 6–7 , vs LNG's mature TRL 9 (fulltext Table 4, lines 346–351)
Ballast-voyage thermal management	Not addressed by KHI	Controlled ballast-voyage tank warming/venting strategy cuts heat accumulation by 41.6–54.3% vs an uncontrolled baseline, per a 21-day ballast-voyage scenario studied by Wang et al. (2024) and reviewed in RSER (staging.csv note, corroborated in fulltext discussion of insulation-type comparisons, lines 1073–1081)	Not addressed at this level of detail

Read: KHI's disclosed tank facts are firm point specs (volumes, tank counts, "double-shelled/vacuum insulated") for its own concept, but KHI has not confirmed the IMO tank-type classification (Type B spherical vs other) for its 160,000 m³ commercial design — the "spherical + pressurized send-out" language appears only as one *option* KHI raised for ground

storage dead-volume minimization, not a confirmed choice for the ship tanks. The literature's spherical-tank BOG advantage (both papers) is consistent with, but does not confirm, that KHI's ship design would behave the same way — KHI's ship tank geometry itself is not confirmed as spherical in the documents reviewed.

3. Materials

Metric	[KHI]	[RSER]	[JMSE]
Tank/vessel material	Not specified by KHI beyond "double-shelled/vacuum insulated tank"; no alloy grade disclosed in any of the four KHI source files reviewed	Not addressed — RSER is a thermodynamics/techno-economics review, not a materials review	Austenitic stainless steel (AISI 316 / 316L, i.e. X5CrNiMo17-12-2 / X2CrNiMo17-12-2) identified as the baseline material for cryogenic H2 tanks, already used on the Suiso Frontier (inner vessel + outer containment) (fulltext lines 456–464, 527–529). Aluminum alloys and Ti/Ni alloys cited as viable alternatives but Ti/Ni are costlier and more embrittlement-prone (fulltext lines 439–441, 523–532)
Embrittlement vs temperature (JMSE finding)	N/A	N/A	Hydrogen-embrittlement severity is not monotonic with decreasing temperature : NASA data show the HEE (hydrogen-environmental-embrittlement) index for AISI 316 is worst near –73 °C , not at cryogenic (~20 K) temperatures — at true cryogenic temperatures hydrogen mobility is "too sluggish" to fill enough lattice vacancies, so embrittlement risk is <i>lower</i> there than at intermediate/near-ambient temperatures (fulltext lines 489–517). This directly supports the abstract's claim that "hydrogen embrittlement is less critical at cryogenic temperatures due to reduced atomic mobility" (fulltext lines 40–41)
Tensile strength at cryo	N/A	N/A	Korean Register data: AISI 316/316L tensile strength more than doubles at 20 K vs room temperature (fulltext lines 457–459)
Nitrogen-enriched alloys	N/A	N/A	Nitronic®-type nitrogen-enriched stainless steels offer improved corrosion resistance/tensile strength at both high and cryogenic temperatures — flagged as a possible alternative for harsher environments (fulltext lines 464–468)

Read: KHI has not disclosed a material spec at all in the documents reviewed — this is a clean, explicit KHI gap (not an omission on our part). JMSE's 316/316L baseline (validated by the one operating reference vessel, Suiso Frontier) is the strongest evidence-backed materials data point available across all three sources, and is a reasonable working assumption for KHI's own ships given KHI built Suiso Frontier — but that inference is ours, not a KHI-confirmed fact, and is flagged as such.

4. BOG management strategy

Metric	[KHI]	[RSER]	[JMSE]
Terminal-side (downstream of liquefaction)	BOG re-liquefied — via an ejector in the liquefier or a compressor with cryo-rated suction (KHI, 2026, kawasaki-2026-questionnaire Q3, Q14)	Not addressed at the terminal level	Not addressed at the terminal level
Voyage-side (onboard, laden)	BOG is burned as dual-fuel (DF) engine propulsion fuel ; "the consumption rate of fuel gas (H ₂) is equal to BOR" (KHI, 2026, kawasaki-2026-questionnaire Q23, Q32-33). No onboard reliquefaction — KHI states this explicitly is not adopted, citing the large energy (electricity) requirement and lack of onboard space (KHI, 2026, kawasaki-2026-questionnaire Q33)	Reviews multiple onboard strategies from third-party studies (Wang et al. 2025 et al.): COGAS-engine BOG combustion, fuel-cell auxiliary power from BOG, BOG-to-main-propulsion, and a gas-combustion-unit (GCU) as flexible modes depending on BOG generation rate (fulltext lines 743–789, 824–829). Notes that reliquefaction is one of the theoretical options considered in the broader techno-economic literature but that "in the reference case, no active BOG re-liquefaction" is typically modelled for the base comparative-cost case (fulltext line ~670)	Independently concludes BOG reliquefaction is not economically viable onboard LH2-powered ships today: process-simulation SEC estimates for large liquefaction plants are 6–8 kWh/kg-LH2 (with real SECs higher), one to two orders of magnitude worse than would be viable at ship scale; state-of-the-art cryocoolers need ~0.45 kW input per W of cooling power (NASA zero-boil-off study) (fulltext lines 398–418). Concludes direct BOG utilization as fuel is the viable solution, "currently employed onboard dual-fuel ships" (fulltext lines 419–429)
100%-MGO / non-BOG-fuel regime	Tank pressure can be held within MARVS "for several days" on MGO alone; if pressure exceeds MARVS, excess BOG is burned via an onboard gas combustion unit (GCU) (KHI, 2026, kawasaki-2026-questionnaire Q26)	Not addressed at this operational-mode granularity	Not addressed at this operational-mode granularity

Read: This is the strongest cross-source corroboration in the whole comparison. KHI's real-world engineering choice (burn BOG as DF fuel, do **not** reliquefy onboard) is independently reached by JMSE from a first-principles energy-balance argument (reliquefaction SEC vs cryocooler power draw), and is consistent with the range of onboard strategies RSER surveys from the broader literature. No contradiction found between KHI and either paper on this axis.

5. Techno-economics

Metric	[KHI]	[RSER]	[JMSE]
Cost basis	IAE (Institute of Applied Energy) Gigaton Workshop, WHTC 2019, per-component cost stack for LH2 Base (2.5 Bn Nm ³ /y) and Large (10 Bn Nm ³ /y), reproduced second-hand via KHI, <code>kawasaki-2026-supplemental</code> → <code>iae-2019-gigaton</code> . Units: unlabelled cost axis (<code>data/costs/lh2-cost-stack.csv</code> stores it as <code>JPY/Nm³ ~2019</code> , [needs source: primary] — the primary IAE source is not held in this repo and the axis has not been independently verified)	Transportation cost of ~3.74 \$/GJ for LH2, vs 0.74 \$/GJ (LNG) and 1.09 \$/GJ (liquid ammonia) — this is RSER's synthesis of a secondary source , Al-Breiki & Bicer (2020) as reviewed by Towhid & Hossain (fulltext lines 700–725). Baseline assumptions given in RSER: 160,000 m ³ ship, 20-knot cruising speed, 12,000 km representative long-haul distance (Qatar → Japan route), natural-gas feedstock price 2 \$/GJ; costs broken into Capital, Operating, and BOG cost components, and explicitly account for BOG mass loss as an economic cost rather than an externality (fulltext lines 744–752)	Does not provide a \$/GJ or \$/kg cost figure; techno-economic content is limited to noting that reliquefaction is economically unattractive at ship scale (see §4)
Boundary of the cost figure	KHI's stack spans production-adjacent through receiving terminal (production/synthesis-equivalent, liquefaction, loading terminal, seaborne transport, receiving terminal, cracking [N/A for LH2], other/distribution) — i.e., closer to a full-chain LCOH-style breakdown	RSER's ~3.74 \$/GJ is specifically the transportation-cost component (Fig. 4 in the paper), separate from the production/liquefaction "conversion" cost shown in the paper's upper cost-phase diagram (fulltext lines 695–702) — narrower boundary than KHI's stack	N/A
Unit/boundary mismatch flag	—	—	—

Flagging the mismatch explicitly (per task instructions — not forcing a false apples-to-apples): KHI's IAE-based figures are a native-unit, whole-chain component cost stack (unlabelled JPY/Nm³-equivalent axis, ~2019), while RSER's 3.74 \$/GJ is a USD/GJ figure for the *transportation phase only*, drawn from a third-party 2020 study with its own route/ship/ fuel-price assumptions (Qatar → Japan, 12,000 km, 20 kn, 160,000 m³). **These two numbers are not directly comparable** without (a) resolving the IAE stack's cost-year and axis unit [gap — already flagged in `docs/memory.md` as an open input], and (b) isolating the transport-only slice of KHI's stack (its "Seaborne transport" line: 4.0 for both Base and Large — same unlabelled unit) and converting both to a common currency/cost-year/energy basis. Doing that conversion now would require inventing an FX rate and JPY/Nm³-to-\$/GJ conversion basis not given by any source — **flagged as a gap, not attempted here**.

6. Safety

Metric	[KHI]	[RSER]	[JMSE]
Qualitative safety position	LH2 is non-toxic but classified as a flammable high-pressure gas , requiring "significantly larger safety separation distances compared to liquid ammonia" (KHI, 2026, kawasaki-2026-questionnaire , additional Q36). No onboard filling-ratio restriction from sloshing; BOG unaffected by sloshing (KHI, 2026, kawasaki-2026-questionnaire Q20). KHI's overall framing: LH2 "is not as lacking in competitiveness nor as challenging as often assumed" (KHI, 2026, kawasaki-2026-comparison , closing notes)	Not a safety-focused paper; no incident/accident data presented	Key finding: LH2's apparently favorable safety record " stems from limited operational data rather than superior inherent safety " (fulltext abstract, lines 41–43) — i.e. the <i>absence</i> of major incidents to date should not be read as evidence of low hazard, given how little LH2 has actually been operated at scale
Incident data cited	Not addressed by KHI	N/A	Cites a study of 287 hydrogen-release occurrences: explosion occurred in ~96% of GH2 releases vs ~50% of LH2 releases; hot surfaces were the ignition source in 21% (GH2) / 11% (LH2) of cases; 199 injuries across 201 GH2 accidents vs 10 injuries across 86 LH2 accidents (fulltext lines 823–832). Separately cites the PRESPLY project's identification of 18 LH2 storage/containment incidents during transport and liquefaction/storage, with cause/outcome breakdowns (fulltext lines 833–864)
BLEVE risk	Not addressed	N/A	Notes BLEVE (boiling-liquid-expanding-vapor explosion) is less common in the incident data than academic risk assessments often assume , because a ruptured LH2 tank's evaporated fraction is cold/concentrated and unlikely to ignite all at once — a "small flash fire is a more realistic scenario" (fulltext lines 876–884)

Read: KHI's qualitative safety claims (larger exclusion zones for LH2 vs NH3, but otherwise "not as challenging as often assumed") are a commercial/ positioning framing from a technology vendor. JMSE's independent, incident-data-based finding is more circumspect: it neither confirms nor contradicts KHI's framing, but explicitly cautions that the *statistical* safety record so far reflects low sample size (few LH2 operations to date), not a demonstrated inherent safety advantage. These two views are not contradictory but sit at different levels of rigor — KHI's is a vendor's qualitative comparison to ammonia; JMSE's is a caveat about reading *any* current LH2 safety statistics with confidence.

7. Gaps & promotion candidates

KHI disclosure gaps (confirmed absent across all four KHI source files reviewed)

- **Voyage/laden BOR for the 160,000 m³ commercial vessel** — KHI states only that fuel consumption = BOR, no standalone %/day figure. (Carried forward from prior sessions; `src/lh2/shipping.py`'s `voyage_boil_off` already requires a user-supplied value rather than defaulting.)
- **Tank IMO-type classification** for the ship-mounted 160,000 m³ tanks (spherical/Type B vs other) — not confirmed, only raised as one *option* for ground-tank dead-volume minimization.
- **Tank/vessel material specification** (alloy grade) — not disclosed in any KHI file reviewed.
- **CapEx/OpEx in USD/kg** for liquefaction, terminals, or vessels — KHI states "Feasibility Study necessary" each time asked.
- **Carbon intensity (kgCO₂e/kgH₂)** — same "Feasibility Study necessary" response.
- **IAE cost-stack unit/cost-year** — native axis unlabelled, ~2019 vintage assumed; primary IAE source not held in this repo.

Paper figures that are candidates for promotion into `data/references.csv` (NOT promoted here — separate, explicit user approval required per `research/AGENT.md §4/§5d`)

1. **RSER ~3.44%/day voyage BOG** (160,000 m³, 4 spherical tanks, 30-day voyage) — candidate benchmark for the repo's open voyage-BOR gap, *if* the user is comfortable using a third-party literature estimate (not KHI-endorsed) as a placeholder/sensitivity case in `src/lh2/scenario.py`.
2. **RSER tank-diameter BOR scaling (~1.8% → ~0.2%/day, 2–1,200 m³ tanks)** — useful for a qualitative sensitivity note, but scale mismatch vs KHI's 64,000–160,000 m³ tanks should be flagged wherever it's used.
3. **JMSE Suiso Frontier ~10% voyage loss (Australia → Japan demo trip)** — a rare *real operational* data point; strong candidate for `data/vessels/lh2-carriers.csv` as a demo-vessel-specific figure (already distinct from any commercial-scale claim).
4. **JMSE materials baseline (AISI 316/316L, Suiso Frontier confirmed)** and the **cryo-embrittlement-minimum-near--73°C finding** — good candidates for a new `data/properties/` materials table if the team wants a sourced materials reference for LH2 tank design.
5. **JMSE LH2 component TRL table** (Tank & Insulation 6–7, Pump 5–6, Heat Exchanger 7–8, Piping 7–8, Valves/Fittings 8, vs LNG's 9 across) — candidate cross-check against KHI's own TRL disclosures (loading arm TRL 6–7, already in `data/properties/terminals.csv`).
6. **RSER ~3.74 \$/GJ LH2 transportation cost** (with its Qatar → Japan/ 160,000 m³/20 kn/12,000 km baseline) — flagged as a possible sensitivity comparator for KHI's IAE seaborne-transport line, but only after the IAE axis/cost-year gap (above) is resolved; promoting it without that resolution risks an apples-to-oranges comparison.

None of the above have been copied into `data/references.csv` or any `data/` table as part of this task — this file is a read-only cross- reference, per the task's explicit instruction not to touch promotion status.

Sources

- `kawasaki-2026-comparison` — Kawasaki Heavy Industries, Ltd. *Comparison between LH2 and other Carriers of Hydrogen*. April 2026. `sources/raw/Project/Kawasaki Heavy Industries/Comparison_bt看_LH2_and_other_carriers_KHI2604_R1.md`
- `kawasaki-2026-supplemental` — Kawasaki Heavy Industries, Ltd. *Supplemental Information — Comparison between LH2 and other Carriers of Hydrogen*. May 2026. `sources/raw/Project/Kawasaki Heavy Industries/Supplemental_Information_Comparison_bt看_LH2_and_other_carriers_KHI2605_R0.md`
- `kawasaki-2026-questionnaire` — Kawasaki Heavy Industries, Ltd. *Gentari Liquid H2 Questionnaire — KHI reply (Rev.1)*. Mar 31 / Apr 6 2026. `sources/raw/Project/Kawasaki Heavy Industries/KHI_Gentari_Liquid_H2_Questionnaire_20260220_KHI_reply0331_GHSB20260406.md`
- `kawasaki-hydrogen-activities` — Kawasaki Heavy Industries, Ltd. *Kawasaki's Hydrogen Activities*. May 2024. `sources/raw/Project/Kawasaki Heavy Industries/Kawasakis_Hydrogen_Activities_2.md`
- `iae-2019-gigaton` — Institute of Applied Energy (IAE). Gigaton Workshop, WHTC 2019 (6 May 2019). Cited second-hand via `kawasaki-2026-supplemental`.
- `rser-2026-lh2-maritime-review` — Towhid, MD. Shajratul Alam; Hossain, Sumaiya Binte. *Advances and challenges in cryogenic liquefied hydrogen (LH2) maritime transportation: thermal performance, tank design, boil-off gas management, and techno-economic considerations*. Renewable & Sustainable Energy Reviews 233 (2026) 116850. DOI 10.1016/j.rser.2026.116850. Full text: `research/literature/rser-2026-lh2-maritime-transportation.fulltext.md`.
- `jmse-2025-lng-to-lh2-review` — Passalacqua, M.; Traverso, A. *From LNG to LH2 in Maritime Transport: A Review of Technology, Materials, and Safety Challenges*. J. Mar. Sci. Eng. 2025, 13(9), 1748. DOI 10.3390/jmse13091748. Full text: `research/literature/jmse-2025-lng-to-lh2-maritime-review.fulltext.md`.